Signature

Tuesday, 4 August, 2026 8:09 PM

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Easy

Tuesday, 4 August, 2026 8:33 PM

Formula: C = (P + k) mod 26 (encrypt) | P = (C − k) mod 26 (decrypt)

Easy

Encrypt the word HAPPY using shift key k = 7.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

C = (P + k) mod 26

H : (7 + 7) mod 26 = 14 : O

A : (0 + 7) mod 26 = 7 : H

P : (15 + 7) mod 26 = 22 : W

Y : (24 + 31) mod 26 = 5 : F

Answer : OHWWF

———————— #

Medium

Tuesday, 4 August, 2026 8:58 PM

Formula: $C = (P + k) \bmod 26$ (encrypt) | $P = (C - k) \bmod 26$ (decrypt)

Medium

Decrypt the ciphertext "WKLV LV IXQ" given shift key k = 3. Then state what makes the Caesar cipher weak against brute-force attacks (hint: how many possible keys exist?).

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

$\therefore W(22) - 3 = 19 \rightarrow T$

$\therefore K(10) - 3 = 7 \rightarrow H$

$\therefore L(11) - 3 = 8 \rightarrow I$

$\therefore V(21) - 3 = 18 \rightarrow S$

$\therefore L(11) - 3 = 8 \rightarrow I$

$\therefore I(8) - 3 = 5 \rightarrow F$

$\therefore X(23) - 3 = 20 \rightarrow U$

$\therefore Q(16) - 3 = 13 \rightarrow N$

Answer: THIS IS FUN

#

Weakness : Only 26 possible keys total, so it is trivial to brute-force by hand.

Hard

Tuesday, 4 August, 2026 9:05 PM

Formula: C = (P + k) mod 26 (encrypt) | P = (C − k) mod 26 (decrypt)

Hard

You are given only the ciphertext " ORWMCQNTNH" and told it is a Caesar cipher of a common English word. Find the key by trying all 25 shifts (brute force) and identify the plaintext.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

"ORWMCQNTNH"

A (0) - 9 = -9 → R

B (1) - 9 = -8 → S

C (2) - 9 = -7 → T

D (3) - 9 = -6 → U

E (4) - 9 = -5 → V

F (5) - 9 = -4 → W

G (6) - 9 = -3 → X

H (7) - 9 = -2 → Y

I (8) - 9 = -1 → Z

J (9) - 9 = 0 → A

K (10) - 9 = 1 → B

L (11) - 9 = 2 → C

M (12) - 9 = 3 → D

N (13) - 9 = 4 → E

O (14) - 9 = 5 → F

P (15) - 9 = 6 → G

Q (16) - 9 = 7 → H

R (17) - 9 = 8 → I

S (18) - 9 = 9 → J

T (19) - 9 = 10 → K

U (20) - 9 = 11 → L

V (21) - 9 = 12 → M

W (22) - 9 = 13 → N

X (23) - 9 = 14 → O

Y (24) - 9 = 15 → P

Z (25) - 9 = 16 → Q

∴ Answer: FIND THE KEY

Easy

Tuesday, 4 August, 2026 9:23 PM

Formula: $C = (aP + b) \bmod 26$ | $P = a^{-1}(C - b) \bmod 26$, where a^{-1} is the modular inverse of a mod 26 (requires gcd(a, 26) = 1).

Easy

Encrypt the single letter K using key a = 5, b = 8.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

K value = 10 → $C = (5 \times 10 + 8) \bmod 26$

$= 58 \bmod 26$

$= 6$

Encryted "K" = G ✱

Medium

Tuesday, 4 August, 2026 9:27 PM

Formula: $C = (aP + b) \bmod 26$ | $P = a^{-1}(C - b) \bmod 26$, where a^{-1} is the modular inverse of a mod 26 (requires gcd(a, 26) = 1).

Medium

Encrypt the word MATH using key a = 7, b = 3. Show each step (numeric value → apply formula → convert back to letter).

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

Encrypt "Math" $C = (aP + b) \bmod 26$

$a = 7, \; b = 3$

$\therefore M(12) = (7 \times 12 + 3) \bmod 26$
$= 87 \bmod 26$
$= 9$ → J

$\therefore A(0) = (7 \times 0 + 3) \bmod 26$
$= 3 \bmod 26$
$= 3$ → D

$\therefore T(19) = (7 \times 19 + 3) \bmod 26$
$= 136 \bmod 26$ → G

$= 6$

$\therefore H(7) = (7 \times 7 + 3) \bmod 26$ } A

$= 52 \bmod 26$

$= 0$

Answer : JDGA 卅

Hard

Tuesday, 4 August, 2026 9:37 PM

Formula: $C = (aP + b) \bmod 26$ | $P = a^{-1}(C - b) \bmod 26$, where a^{-1} is the modular inverse of a mod 26 (requires gcd(a, 26) = 1).

Hard

You intercept ciphertext " IMZZQLT", encrypted with an affine cipher where a = 9. You know the first plaintext letter is F. Find b, then decrypt the full message. (Recall: you'll need the modular inverse of 9 mod 26.)

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

"IMZZQLT", a = 9, first plaintext = F

$P = a^{-1}(C - b) \bmod 26$

∴ find a^{-1}

- $9x \equiv 1 \pmod{26}$
- $9 \times 3 = 27 \equiv 1$
- $a^{-1} = 3$

∴ find b

I = 8, F = 5

- $8 = (9 \times 5 + b) \bmod 26$
- $(45 + b) \bmod 26$
- $(19 + b) \bmod 26$
- $b = 8 - 19 \equiv -11 \equiv 15$

∴ decrypt with $p = 3(C - 15) \bmod 26$

→ I(8) : $3(-7) = -21 \equiv 5$ → F

→ M(12) : $3(-3) = -9 \equiv 17$ → R

→ Z(25) : $3(10) = 30 \equiv 4$ → E } x2

→ Q (16): $3(1) = 3 \longrightarrow D$

→ L (11): $3(-4) = -12 \equiv 14 \longrightarrow O$

→ T (19): $3(4) = 12 \longrightarrow M$

Answer: FREEDOM

Easy

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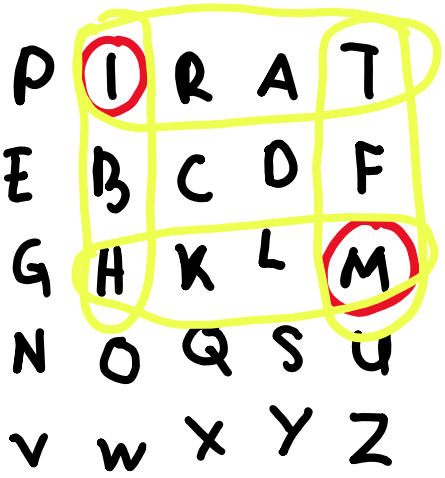
Rules: Build a 5×5 key square from a keyword (drop duplicate letters, merge I/J). Split plaintext into digraphs, inserting a filler (commonly X) between repeated letters in a pair and padding an odd final letter. Encrypt pairs by: same row → shift right; same column → shift down; rectangle → swap columns.

Easy

Using keyword PIRATE, construct the 5×5 key square, then encrypt the digraph TH.

Answer :

keyword "PIRATE", encrypt TH



T = Row 0, Col 4

H = Row 2, Col 1

different row/col → Rectangle rule → grid[0][1] = I, grid[2][4] = M

∴ Answer = IM #

Medium

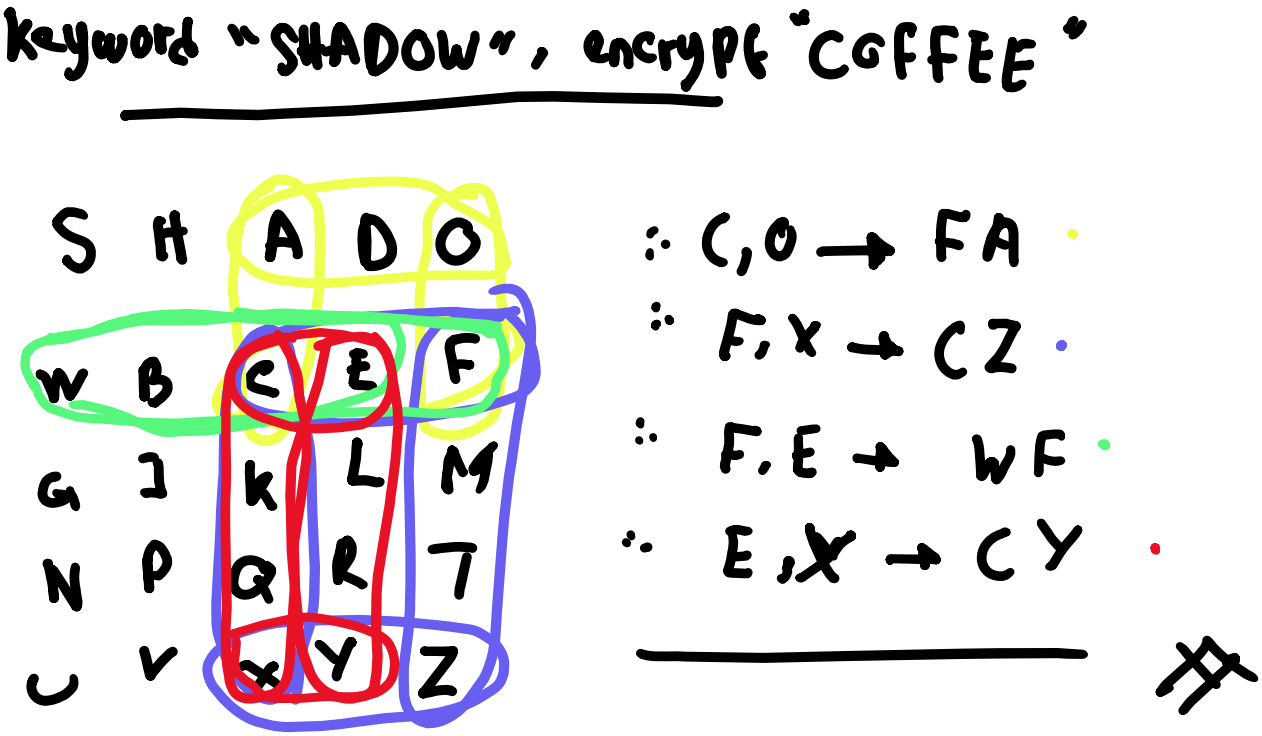
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Rules: Build a 5×5 key square from a keyword (drop duplicate letters, merge I/J). Split plaintext into digraphs, inserting a filler (commonly X) between repeated letters in a pair and padding an odd final letter. Encrypt pairs by: same row → shift right; same column → shift down; rectangle → swap columns.

Medium

Using keyword SHADOW, encrypt the word COFFEE (handle the double-F using the filler rule).

Answer :



Hard

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Hard

Using keyword GALAXY, encrypt the phrase SEND HELP NOW (remove spaces first, apply padding rules as needed), then decrypt your own ciphertext back to confirm it matches the original.

Answer :

I) Medium

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How it works (step by step):

1. Convert each letter of the plaintext and the keyword to a number: A=0, B=1, C=2, … Z=25.
2. Repeat the keyword underneath the plaintext until it's the same length (ignore spaces in the plaintext when doing this).
3. For each position, add the plaintext letter's number to the keyword letter's number, then take the remainder when divided by 26 (wrap back to A if you go past Z).
4. Convert the result back to a letter and that's your ciphertext letter.

Medium

Encrypt the word LEMON using the keyword CAT. Write out the letter-by-letter number conversion, repeat the keyword to match the length, add and wrap around, then convert back to letters.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

Encrypt "LEMON", key "CAT"

∴ L + C = 11 + 2 → 11 + 2 = 13 → 13 mod 26 = 13 → N

∴ E + A = 4 + 0 → 4 + 2 = 4 → 4 mod 26 = 4 → E

∴ M + T = 12 + 19 → 12 + 19 = 31 → 31 mod 26 = 5 → F

∴ O + C = 14 + 2 = 14 + 2 = 16 → 16 mod 26 = 16 → Q

∴ N + A = 13 + 0 = 13 + 0 = 13 → 13 mod 26 = 13 → N

∴ Answer = NEFQN #

2) Medium

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Medium

Encrypt the phrase GOOD LUCK using the keyword SUN (ignore the space, repeat the keyword across all 8 letters). Show your working in a table with columns: Plaintext letter, Plaintext number, Key letter, Key number, Sum mod 26, Ciphertext letter.

Answer :

Encrypt "GOOD LUCK", Key "SUN"

∴ G + S = 6 + 18 → 6+18 = 24 → 24 mod 26 = 24 → Y

∴ O + U = 14 + 20 → 14 + 20 = 34 → 34 mod 26 = 8 → I

∴ O + N = 14 + 13 → 14 + 13 = 27 → 27 mod 26 = 1 → B

∴ O + S = 3 + 18 → 3 + 18 = 21 → 21 mod 26 = 21 → V

∴ L + U = 11 + 20 → 11 + 20 = 31 → 31 mod 26 = 5 → F

∴ U + N = 20 + 13 → 20 + 13 = 33 → 33 mod 26 = 7 → H

∴ C + S = 2 + 18 → 2 + 18 = 20 → 20 mod 26 = 20 → U

∴ K + U = 10 + 20 → 10 + 20 = 30 → 30 mod 26 = 4 → E

∴ Answer = YIBVFHUE #

3) Medium

Tuesday, 4 August, 2026 10:35 PM

Medium

You are given the ciphertext YECDCCIM QRGEXI and told the keyword used was MERDEKA. Decrypt it back to the original plaintext by subtracting the keyword numbers from the ciphertext numbers (wrap around if needed).

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

Decrypt "YECDCCIM QRGEXI", key "MERDEKA"

- Y (24) − M (12) = 12 → M
- E (4) − E (4) = 0 → A
- C (2) − R (17) = −15 ≡ 11 → L
- D (3) − D (3) = 0 → A
- C (2) − E (4) = −2 ≡ 24 → Y
- C (2) − K (10) = −8 ≡ 18 → S
- I (8) − A (0) = 8 → I
- M (12) − M (12) = 0 → A

- Q (16) − E (4) = 12 → M
- R (17) − R (17) = 0 → A
- G (6) − D (3) = 3 → D
- E (4) − E (3) = 0 → A
- X (23) − K (10) = 13 → N
- I (8) − A (0) = 8 → I

Answer : MALAYSIA MADANI

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Easy (2x2)

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2×2 Matrix

Formula: Break plaintext into pairs of numeric vectors, $C = K \cdot P \bmod 26$; decrypt with $K^{-1} \bmod 26$.

Easy

Using key matrix K = [[3,3],[2,5]], encrypt the digraph HI.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

K = [[3,3],[2,5]], encrypt HI

P = [7,8]

∴ Row 1 : 3(7)+3(8) = 45 mod 26

= 19 → T

∴ Row 2 : 2(7) + 5(8) = 54 mod 26

= 2 → C

∴ Answer = TC ✱

Medium (2x2)

Tuesday, 4 August, 2026 10:53 PM

2×2 Matrix

Formula: Break plaintext into pairs of numeric vectors, $C = K \cdot P \bmod 26$; decrypt with $K^{-1} \bmod 26$.

Medium

Using key matrix K = [[5,8],[17,3]], encrypt the word PAY (pad with X if needed to make an even length).

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

K = [[5,8],[17,3]], encrypt "PAY" (pad → PAYX) to make even length

PA : Row 1 = 5(15) + 8(0) = 75 mod 26
= 23 → X
Row 2 = 17(15) + 3(0) = 255 mod 26
= 21 → V
} XV

YX : Row 1 = 5(24) + 8(23) = 304 mod 26
= 18 → S
Row 2 = 17(24) + 3(23) = 477 mod 26
= 9 → J
} SJ

∴ Answer : XVSJ #

Hard (2x2)

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Hard

Given key matrix K = [[3,3],[2,5]], first verify it is invertible mod 26 (compute the determinant and check gcd(det, 26) = 1), find K^{-1} mod 26, then decrypt the ciphertext HIAT.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Answer :

K = [[3,3],[2,5]], find K^{-1}, decrypt HIAT

1. det (K) = 3(5) − 3(2) = 9
   gcd (9, 26) = 1 → invertible ✓

2. 9^{-1}: 9 × 3 = 27 ≡ 1 → 9^{-1} = 3

3. adj (K) = [[5, −3], [−2, 3]]; K^{-1} = 3 × adj mod 26
   = [[15, 17], [20, 9]]

4. Decrypt:

HI : C = [7, 8]: Row 1 = 15(7) + 17(8)
= 241 mod 26
= 7 → H
Row 2 = 20(7) + 9(8)
= 212 mod 26
= 4 E
} HE

∴ Answer : HELP ✓

AT : C = [0, 19]: ROW 1 = 17(19)
= 323 mod 26
= 11 → L
Row 2 = 9(19)
= 171 mod 26
= 15 → P
} LP

Easy (3x3)

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Key matrix used for Easy and Medium questions:

```
K = [ 6  24  1 ]     [13  16 10 ]     [20  17 15 ]
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Easy

Using the 3×3 key matrix above, encrypt the trigraph ACT.

Answer :

Encrypt "ACT", P [0, 2, 19]

- Row 1 : 24 (2) + 1 (19) = 67 mod 26
  = 15 → P
- Row 2 : 16 (2) + 10 (19) = 222 mod 26
  = 14 → O
- Row 3 : 17 (2) + 15 (19) = 319 mod 26
  = 7 → H

Answer : POH